

HAT-P-32b

R-1228

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_211013 · created 2026-07-31T09:31:00+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459500.9164 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1778 +/- 0.0042
Transit depth (Rp/Rs)^2	3.16 +/- 0.15 [%]
Orbital Inclination (inc)	88.66 +/- 1.06
Airmass coefficient 1 (a1)	1.42 +/- 0.012
Airmass coefficient 2 (a2)	-0.3324 +/- 0.0093
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	None
Optimal Aperture	5.7
Optimal Annulus	25.21
Transit Duration (day)	0.1332 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1778 $\pm$ 0.0042 vs 0.1489 (deviation 6.25 σ)
Duration fit vs published	0.1332 d vs 0.1304 d (deviation 2.3 σ)
Mid-time O-C	0.2 min
Depth SNR	38.4
Residual scatter	0.82 %

Light curve

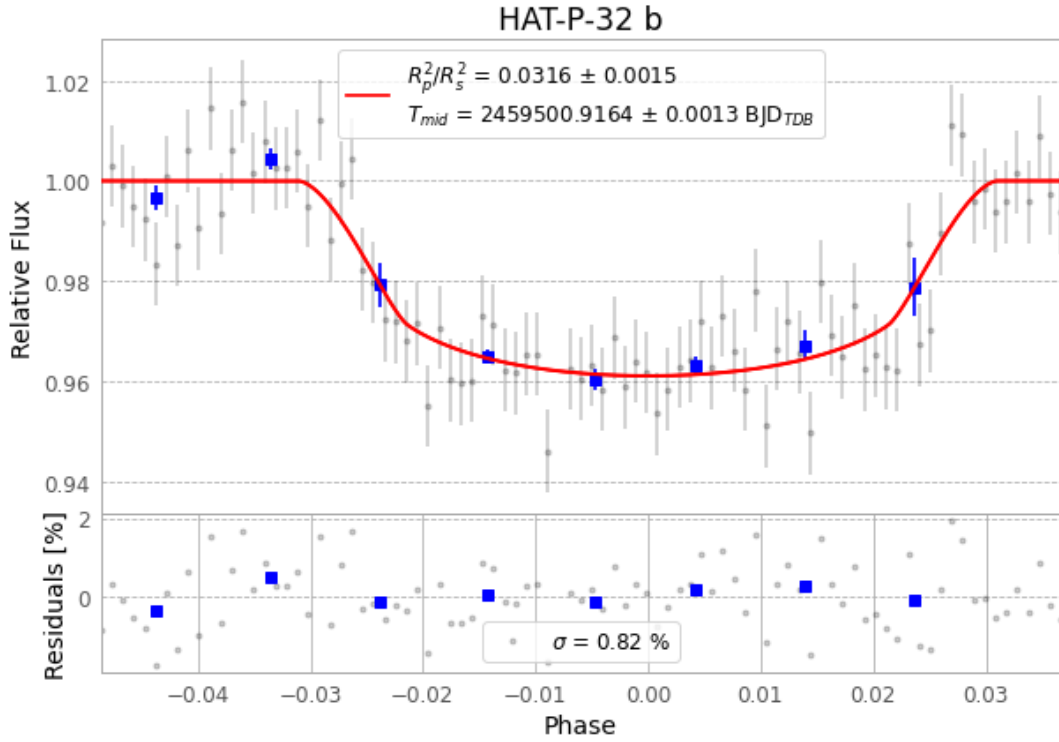


Figure 1 — FinalLightCurve_HAT-P-32 b_2021-10-13.png (SHA-256 803b4fd926192550...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [437, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3ccae1cc87878336...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

88 FITS frames (58 MB), HATP-32211013072112.FITS → HATP-32211013114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-211013.log (59563b312f11...), FinalLightCurve_HAT-P-32 b_2021-10-13.png (803b4fd92619...), FinalLightCurve_HAT-P-32 b_2021-10-13.csv (0cf26b25030f...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 09:31Z.